Technical Brief

# Aragonite vs Limestone: a comparative technical analysis

Crystal system, morphology and evaluation considerations for natural oolitic aragonite and conventional calcite limestone.

## Summary

Aragonite and calcite share the formula $CaCO_3$, but they are distinct mineral polymorphs. Limestone is composed primarily of calcite, while AragoCor supplies natural oolitic aragonite with an orthorhombic crystal system and rounded, concentrically layered grains. These differences can influence physical properties and process behavior; suitability still depends on grade, formulation and operating conditions.

## Side-by-side: aragonite vs calcite

| Property | AragoCor Aragonite | Calcite / Limestone | Industrial impact |
|---|---|---|---|
| Crystal system | Orthorhombic | Trigonal / rhombohedral | Distinct lattice and physical properties |
| Density (g/cm³) | ~2.93 | ~2.71 | Different material density |
| Relative solubility | Generally higher | Generally lower | pH, temperature and particle size also matter |
| Typical particle morphology | Rounded, oolitic grains | Deposit- and milling-dependent | Flow and packing should be tested by grade |
| Purity | ≥97.5% $CaCO_3$ typical | Product-dependent | Confirm with current documentation |
| Origin | Natural Bahamian deposit | Deposit-dependent | Source and logistics differ |
| Iron content | ≤0.04% $Fe_2O_3$ typical | Product-dependent | Confirm for clarity-sensitive formulations |

## Why the difference matters

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|---|---|
| Reactivity<br>Compare dissolution or thermal behavior under the intended pH, temperature and formulation conditions. | Purity<br>Review the current specification and available lot-specific chemistry for the grade offered. |
| Morphology<br>Evaluate how rounded oolitic grains affect flow, packing, blending or dispersion in the process. | Source<br>Consider mineral origin, processing, packaging, freight and chain-of-custody requirements. |

## Application considerations

Aragonite may be evaluated as an alternative to conventional limestone. Because performance is application-specific, AragoCor recommends a documented trial before full specification. A technical specialist can help identify a starting material, relevant documentation and practical trial questions.

## Conclusion

Natural oolitic aragonite and calcite limestone are different crystal forms of calcium carbonate. That distinction is a useful starting point for comparison, but current documentation and application testing should drive the final selection.

This brief summarizes representative properties for comparative and educational purposes. Values are not guaranteed; request current specifications and available lot-specific documentation. © AragoCor Minerals LLC. info@aragocorminerals.com · +1 (209) 487-0110